

RESEARCH PROPOSAL REPORT

Detecting Fish Hunger Levels by Monitoring and Classifying Their Underwater Swimming Patterns

A Low-Cost Camera System for Need-Based Fish Feeding in Bangladesh

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Abstract

Fish farmers need to give the right amount of feed at the right time. High feed costs are a problem for farmers in Bangladesh. Giving too much feed can waste money. It can also harm water quality. A fixed feeding time does not always show whether fish are ready to eat.

This study will test a low-cost system that watches fish through an underwater camera. The system will track fish movement. It will measure swimming speed, turns, position in the image, and how closely fish swim together. A computer model will use these patterns to estimate three hunger levels: Low, Moderate, and High. These levels will be estimates of feeding readiness. They will not be direct measurements of hunger inside a fish.

The project will collect underwater video for model training and testing. It will start with tilapia. It will include pangasius if suitable sites and resources are available. The team will compare simple models with models that learn from short video sequences. A feeding trial will compare camera-guided feeding with normal fixed-time feeding. The team will measure feed use, fish growth, survival, water quality, and costs.

The expected outputs are a working prototype, a labeled video dataset, and a simple guide for farmers. Any feed savings must be measured. Better growth, cleaner water, and higher profit are possible benefits, not guaranteed results.

Keywords: fish hunger; underwater video; fish movement; low-cost camera; need-based feeding; Bangladesh.

1 Introduction and Background

Fish farming is a source of food and income in Bangladesh. However, farmers face many problems. The study by Swarna, Ali, and Galib describes problems faced by seven groups in the freshwater fisheries sector. These include fish farmers, traders, researchers, and consumers [1]. A camera system will not solve all these problems. This proposal focuses on one task: better feeding decisions.

Feed cost is a useful reason to study this task. Fagun and colleagues studied a small group of fish farmers in Habiganj. High feed cost was the most common constraint reported in that study [2]. This was a small local study. Its findings should not be treated as a survey of every farmer in Bangladesh.

Feed is an important farm expense. The team will check actual feed costs at the study farms. Any savings from the system must be measured in the feeding trial.

The idea is simple. A camera records fish. A computer looks for movement patterns linked to feeding readiness. The farmer gets a clear message. The farmer remains in control of feeding.

2 Problem Statement

The proposal addresses four main problems.

1. **Feeding decisions may be uncertain.** A fixed schedule does not measure the current feeding response of fish.
2. **Extra feed may be wasted.** Uneaten feed can add waste to the water. Poor feeding can also reduce farm profit.
3. **A system must be affordable and easy to use.** This project will test ordinary camera parts and a small local computer. Their real cost and usefulness must be checked.

4. **Local data are needed.** A model trained on another species or farm may not work well in local ponds. This project will build data for the selected fish and sites. It does not claim that no related dataset exists anywhere.

3 Related Research and Research Gap

3.1 What earlier studies show

Swarna and colleagues show that fisheries problems differ across stakeholder groups [1]. Their study gives local context. It does not test camera-based hunger detection.

Fagun and colleagues describe fish farming and farmer living conditions in a rural area of Bangladesh [2]. Their work helps explain why a practical feeding tool is worth studying. It does not prove that the proposed system will save money.

Måløy, Aamodt, and Misimi used underwater video to recognize feeding and non-feeding behavior in salmon. Their model combined image information with movement over time [3]. This supports the use of video for feeding behavior research. It does not prove that the same model will classify three hunger levels in tilapia or pangasius.

3.2 What this project will add

This project will test the idea with selected local fish and farm conditions. It will connect movement clips with observed feeding response. It will compare simple and more complex models. It will also test whether using the model helps feeding in practice. The main contribution is this local test, not the claim that video-based fish feeding research is new.

4 Research Aim and Objectives

4.1 Main aim

The aim is to build and test a low-cost camera system that helps farmers decide when to feed fish.

4.2 Specific objectives

The study has the following objectives.

1. Build a camera and local-computer prototype.
2. Detect and track fish. Check detection quality and tracking errors.
3. Classify Low, Moderate, and High feeding readiness. Test the model on data it has not seen before.
4. Collect usable local video and label selected short clips.
5. Compare camera-guided feeding with normal fixed-time feeding and analyze the findings.
6. Prepare a simple operating guide and record the full system cost.

5 Research Questions and Hypotheses

5.1 Research questions

1. Which movement cue best predicts feeding readiness: speed, turns, vertical image position, or group spacing?
2. Does a model that uses movement over time work better than a simpler model?
3. Can camera-guided feeding reduce feed use or improve feed conversion without harming growth?
4. Can the system give useful alerts under the tested local conditions?

5.2 Hypotheses to test

H1: Movement patterns will help predict the three feeding-readiness levels better than a simple time-since-feeding rule.

H2: A model that uses a short sequence of movement will perform better than a model that uses only one image.

H3: Camera-guided feeding will improve feed conversion or reduce feed use compared with the control method.

Growth safety: Camera-guided feeding must not harm fish growth or survival. A fisheries adviser will review the results and fish care plan.

6 Proposed System

6.1 Main parts

- An underwater camera, or a camera in a safe waterproof case.
- A fixed mount that keeps the view steady.
- An existing laptop for early testing and model training.
- A small local computer for testing on-site use.
- Storage, a safe power supply, and backup power where possible.
- Water test tools and a dissolved oxygen meter for safety checks.
- A simple display with a hunger estimate and a suggested action.

The first version will advise the farmer. It will not control an automatic feeder. The display will use simple words. A Bangla display will be tested with users. The goal is local processing without a constant internet connection.

6.2 System flow

Figure 1 shows the main steps of the proposed system.

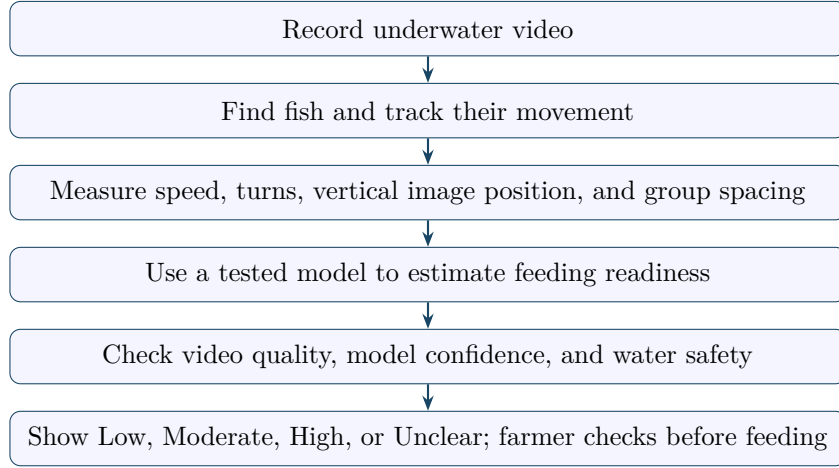


Figure 1: Proposed system workflow.

7 Research Methodology

7.1 Study design and scope

The work will have four stages: pilot testing, data collection, model testing, and a feeding trial. The study will start with tilapia in tanks or ponds where fish can be seen clearly. The final location will depend on permission, fish care, camera access, and available funds. No farm partnership is confirmed in this report.

Pangasius will be a second dataset species if access is possible. The first feeding trial will use one species and one agreed production system. Mixing species across treatment groups could make the comparison unfair. The study will not claim to represent all fish farms in Bangladesh.

7.2 Pilot testing

The team will test the camera angle, lighting, recording quality, and safe mounting. A fisheries adviser will help set normal feeding intervals and feeding-response scores. The pilot will also estimate how much growth and feed conversion vary between units. These estimates will guide the final trial size.

The team will not assume that faster swimming always means greater hunger. It will check movement together with feeding response, water conditions, and fish health.

7.3 Video collection

The team will collect usable recordings. Video will cover normal periods before feeding, during feeding, and after feeding. It will include different days and normal changes in light and water clarity.

Each recording will have a log. The log will include the unit code, species, date, time, last feeding time, feed amount, fish size, and water readings. Unclear video will be marked. It will not be forced into a hunger class.

7.4 How clips will be labeled

A label is a name given to a video clip. The proposed labels describe feeding readiness, not a direct biological measurement of hunger. Time since the last meal will be background information. It will not be the only basis for a label.

The team will record a short clip before a normal feeding event. It will then record the response to a small, measured part of the planned meal. This feed will count toward the daily ration. Labels will combine that response with two trained reviewers' scores and the feeding log.

Table 1: Proposed labeling guide. Exact scoring rules will be set in the pilot.

Label	Evidence used by reviewers
Low	Weak interest in the planned feed, with normal water and health checks.
Moderate	Some interest in feed, but a less active response than the High group.
High	A clear, active response to feed, supported by both reviewers.
Unclear	Poor visibility, reviewer disagreement, or a possible water or health problem.

Reviewers will use a written guide. Their agreement will be reported. Disputed clips will be reviewed or kept as uncertain. Only the pre-feeding clip will be used to predict pre-feeding readiness. Frames showing the later feed response will not be given to that predictor. This avoids letting the model see the evidence used to make its label.

7.5 Detecting and tracking fish

A detector will find fish in each usable frame. A tracker will link fish positions across nearby frames. The team will check examples by hand. It will record missed fish, false detections, and identity changes. When fish overlap, the system may lose a track. A short broken track will not be treated as a complete path.

7.6 Movement features

The team will summarize movement in short video clips. Camera position and visible area will

Table 2: Movement cues and their limits.

Feature	Simple measurement
Swimming speed	Distance moved in the image per second. Use body lengths per second where possible.
Turning frequency	Number of clear direction changes in a time window. The pilot will set the angle rule.
Vertical position	Height in the camera image. This is not true water depth unless the view is calibrated.
Group spacing	Average visible distance between fish, or visible fish count in a fixed image area. This does not measure the whole pond.

be kept as steady as possible. The team will record changes in camera setup. It will compare the same feature definitions across recordings.

8 Model Training and Testing

8.1 Models to compare

The study will start with a simple rule based on time since feeding. It will then compare that rule with a feature-based model, such as a random forest. More complex options will include:

- **CNN:** a convolutional neural network that learns patterns from images.
- **LSTM:** a long short-term memory model that learns changes across a sequence of movement measurements.
- **CNN–LSTM:** a model that combines image patterns with changes over time.

Underwater sequence models have been used in salmon feeding research [3]. This study will test whether a similar approach is useful here. A more complex model will not be chosen only because it is more advanced. It must give a useful gain in accuracy and still run on the test device.

8.2 Keeping the test fair

The team will assign whole pond or tank recording groups to training, validation, and test sets. It will also separate recording dates and sessions. Nearby clips from the same event will stay together. This reduces data leakage, which happens when the test data are too similar to data the model has already seen.

The exact split will depend on the number of independent sites. The test set will remain unused during model selection. If there are too few sites to test a new farm, the team will state that limit clearly. The feeding-trial model will be fixed before the trial starts.

8.3 Main test measures

- **Precision:** how often a predicted class is correct.
- **Recall:** how many true cases of a class the model finds.
- **Macro-F1:** a combined precision and recall score averaged equally across the three classes.
- **Confusion table:** a table showing which hunger levels the model mixes up.
- **Device performance:** time per prediction, power use, and hours of reliable operation.

Scores will be reported for each class, not only as one overall accuracy. The team will also report how often the model refuses to give an estimate. A system that rejects most videos should not appear successful only because it performs well on a few clear clips.

To find useful movement cues, the team will remove one cue at a time and test again. It will also shuffle each cue to see how much the score changes. These checks will show which cues help prediction. They will not prove that a cue causes hunger.

9 Feeding Decisions and Safety

The system will not increase feed without a daily limit. Low dissolved oxygen, unusual behavior, or possible disease will trigger a safety check. A fisheries adviser will set the water limits and response plan before the trial. If water safety is not known, the system will not give a confident feeding recommendation.

Table 3: Proposed advisory actions. All actions remain within the fish care plan.

Output	Farmer action
Low	Recheck fish and water. Delay or reduce the next feed portion only within the approved plan.
Moderate	Continue monitoring. Give an allowed small portion if the feeding plan supports it.
High	Check water safety. Give a measured portion within the daily feed limit.
Unclear	Do not use the estimate. Follow the agreed backup feeding plan.

The farmer can override an alert. Every override and feed amount will be logged. The project will not use long fasting or harmful water conditions to create labels.

10 Field Experiment

10.1 Treatment and control groups

The trial will use independent tanks or ponds arranged in matched pairs. Each pair will have similar fish size, stocking density, water source, and unit size. One unit in each pair will be randomly assigned to each group.

- **Treatment group:** camera-guided feeding, with farmer checks and approved ration limits.
- **Control group:** fixed-time feeding based on fish biomass and the agreed normal feeding plan.

Both groups will use the same feed type and receive the same care. The team will record the actual feed given, not just the suggested amount. Tanks sharing one water loop may not be independent. The design must account for that before random assignment.

The pilot will help the team decide how many independent units are needed. If the study is too small for firm conclusions, the team will describe it as an early test.

10.2 Data to collect

Feed use and feeding actions will be recorded daily. Oxygen and temperature will be checked at agreed daily times and around feeding. Fish behavior and deaths will be logged daily. Fish weight will be sampled regularly with safe handling. The same method will be used in both groups.

The team will measure pH, water clarity, and ammonia on a set schedule. It will also record labor time, camera failures, power use, maintenance, and equipment costs. Start and end fish counts and biomass will be measured as accurately as possible.

10.3 Analysis plan

The main unit of analysis will be the tank or pond, not each fish or video frame. The team will compare paired units and allow for the matched design. It will report differences, uncertainty ranges, and the number of independent units. Repeated water readings will not be treated as independent ponds.

Lower feed use will count as a benefit only if growth and survival remain acceptable. No savings will be claimed before the trial is complete.

11 Risks, Ethics, and Limitations

11.1 Main risks and responses

Table 4: Key risks in the proposed work.

Risk	Planned response
Cloudy water or poor light	Check visibility. Mark unclear clips and use the backup feeding plan.
Fish overlap or leave the view	Check tracking quality. Avoid claims about fish outside the image.
Illness or low oxygen changes movement	Use water and health checks. Do not call every movement change hunger.
Camera or power failure	Inspect equipment, log failures, and keep a normal feeding backup.
Weak or uneven labels	Use two reviewers and report disagreement and class counts.
Small sample size	Check power after the pilot and state uncertainty openly.
High ownership cost	Measure maintenance, power, and replacement costs; ask farmers about usefulness.

11.2 Fish care and participant privacy

The team will obtain the required university review and site permission before work starts. A fisheries adviser will review feed limits, fish handling, oxygen checks, and stopping rules. Fish care will take priority over data collection. The trial will stop or change if there is unsafe water, unusual mortality, or repeated growth concerns.

Farmers will receive a simple explanation of the study. Interviews and feedback will be voluntary. Public records will use site codes instead of personal details. The team will seek consent before sharing video. Faces, voices, and private farm information will be removed where needed.

11.3 Limits of the study

A single camera sees only part of a tank or pond. Image position is not the same as true depth. Movement can change because of stress, light, temperature, or illness. Hunger labels will therefore remain imperfect estimates.

The model may fail on a new species or farm. The team will report the conditions it tested and where it failed. The study will not promise year-round performance or national-scale benefits from one trial.

12 Expected Outcomes and Social Value

The project aims to produce:

1. A working camera-based feeding-readiness prototype.

2. A dataset with selected labeled clips and clear recording notes.
3. A comparison of simple and sequence-based models.
4. Evidence about which movement cues help prediction.
5. A fair comparison of feed use, growth, survival, water quality, and costs.
6. A simple guide explaining setup, safe use, and system limits.

Public data release will depend on consent and licensing. The research may help reduce unnecessary feeding if the trial supports that result. It may also help farmers keep better feeding records. Better water quality and higher profit are possible wider benefits. They must be measured rather than assumed.

A useful result may also be negative. For example, the study may show that a camera is not reliable in very cloudy ponds. Reporting such limits will help later researchers avoid weak designs.

13 Conclusion

This proposal will test whether underwater fish movement can support feeding decisions. It will use a low-cost camera setup, clear labels, and a fair model test. It will then compare camera-guided feeding with a normal feeding plan.

The study is designed to protect fish growth and care. It will not claim savings without evidence. Its value will depend on how well the system works, how much it costs, and whether farmers find it useful.

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